

Colour evidence that the companion of a quasar is itself a quasar at the same redshift

Method note for `qso_pcolor`

September 26, 2026

Abstract

A quasar has a spectroscopic redshift z_0 ; a companion a few arcseconds away has broadband photometry and nothing else. We want to know how strongly the companion’s photometry supports the hypothesis that it is a quasar at $z \simeq z_0$. This note sets out the single model implemented in `qso_pcolor`. It accepts any combination of 41 bands from SDSS, Legacy Surveys DR9 (including its forced unWISE $W1$, $W2$), AllWISE, Pan-STARRS1, NSC, SkyMapper and VHS, including infrared-only input. Quasars and the field population are each described by an extreme-deconvolution mixture over native band luminosities, conditioned on one measured reference band; missing bands are marginalised exactly. Surface densities per unit reference luminosity, built separately for each reference band, turn the two densities into intensities for four hypotheses: a quasar at z_0 , a quasar at another redshift, the modelled field, and a broad “unmodelled” share of the field that stops Gaussian tails from deciding. A narrow redshift window enters as a multiplicative constant, which defines the window-free ranking statistic R . On 50,746 companions of DESI quasars that have spectra of their own, Legacy $g, r, z, W1, W2$ photometry ranks a same-redshift quasar above a wrong-redshift one with held-out AUC 0.828, and separates quasars from stars and galaxies with AUC 0.982 and 0.961; the same-redshift probability is low by a factor that falls from 9.8 at 3–5'' to 3.3 at 20–30'', the signature of quasar clustering that the scorer deliberately omits. Quasars just outside a $\pm 3000 \text{ km s}^{-1}$ window cannot be distinguished from those inside it. The model replaces an earlier Legacy-only model (Appendix C): on identical inputs it ranks redshift better (0.828 against 0.815) and rejects stars slightly less well (0.982 against 0.988). Across all 127 survey combinations, reserved quasars are separated from reserved field sources with median AUC 0.996.

The method in plain words

An orientation before the formalism. Everything asserted here is developed, and qualified, in the sections that follow.

The question, and what is in scope. We have a quasar whose redshift has been measured from a spectrum, and a neighbour a few arcseconds away for which we have only photometry — a brightness in each of a handful of filters. We want to know whether that neighbour is also a quasar, at the same redshift. The pipeline scores only companions far enough from the primary that the survey photometry treats them as separate sources; closer pairs need image-level work that is not attempted here (§12).

Whatever photometry exists. A companion may have been measured by some of seven surveys and not others. The model takes every usable measurement it is given, from any subset of the seven, and integrates over the rest. Filters from different surveys are never treated as the same filter, and the same survey north and south of the Legacy Surveys’ dividing line is treated as two systems.

How we learn what quasars look like. We take spectroscopically confirmed quasars from DESI DR1 and SDSS DR16Q, draw an equal number in every redshift interval of width 0.1, and attach whatever the seven surveys measured for each of them. A fifth of the sky is held back before anything is fitted. For each of a set of overlapping redshift slices we fit the joint distribution of the brightnesses in all bands, and blend neighbouring slices according to where a given redshift falls. At scoring time the distribution is conditioned on one measured band — the reference — so the question it answers is: *if this object is a quasar at redshift z , and is this bright in the reference band, how likely are its brightnesses in the other bands?*

That sample reflects the DESI and SDSS target selections, which are themselves built on these colours. It is not an unbiased sample of quasars in general.

Why measurement error is handled explicitly. Every measurement carries an uncertainty. Fitting the observed cloud would give a locus broader than the true one and would make unusual colours look less unusual than they are. So we fit the *underlying* distribution, using each training object’s own errors, and add each candidate’s own errors back when scoring it. A noisy candidate is compared against an appropriately blurred locus.

What we use for the competing population. Not stars alone. We take every source detected by any of the seven surveys in two dozen small fields spread over the sky, remove the known quasars, and model all of it the same way: stars, faint galaxies, blends, whatever the surveys contain. A chance neighbour is drawn from everything that is there.

How the two are compared. Two distributions of brightness give the relative evidence — reported — but not a probability, because that also depends on how many objects of each kind exist at this brightness. So each is multiplied by a surface density, per unit brightness in the reference band: quasars per square degree per unit redshift, and field sources per square degree. Both are counted only inside the footprint of the survey that supplies the reference band, because a band can be missing either because the object is outside the survey or because it is too faint, and only the second says anything about the object. The result is an expected-count density for each of four possibilities:

- a quasar at the primary’s redshift;
- a quasar at some *other* redshift;
- anything else in the imaging;
- something neither model describes.

The second is not a refinement. A quasar far behind the primary looks nothing like a star, so a quasar-versus-star comparison would count it as evidence for a companion; for a pair search that is the main way of being wrong. The fourth exists because both models are sums of Gaussians. Far from everything either was fitted to, their ratio is decided by whichever Gaussian tail happens to be wider, and an artefact would be called a certain quasar. A small, fitted share of the field with a very broad distribution takes those objects instead.

Why the answer is a ranking rather than a verdict. Colours locate a quasar in redshift only loosely — typically to ± 0.6 — while a $\pm 2000 \text{ km s}^{-1}$ pair window at $z = 1.8$ spans 0.037. Only a small part of any candidate’s redshift probability can fall inside the window, so the probability is small even for a strong candidate. We therefore report a statistic that the width of the window does not affect, in the narrow limit, and convert it to a probability only for a window the user declares. Candidates are ordered by that statistic; spectroscopy decides.

What the test against the truth shows. About 50,000 objects lie within $30''$ of a known quasar *and* have a DESI spectrum of their own. Scored from Legacy photometry alone, a genuine same-redshift partner outranks a random wrong-redshift quasar 83% of the time, and fewer than one star in a thousand scores as high as a typical same-redshift quasar. Quasars just outside the velocity window are indistinguishable from those just inside. And the probability the scorer reports is too small — by about three times for wide pairs and ten times for the closest — because it treats the companion as a random nearby object, while real quasars cluster. The factor is that clustering.

1 The question, and the question that can be answered

The scientific target is physical quasar pairs. One member has a spectrum and the other does not, and obtaining the second spectrum is expensive. A method that ranks companions by how quasar-like their photometry is *at the primary’s redshift* decides where telescope time goes.

A likelihood is not a probability. $p(\mathbf{u} \mid Q, z_0)$ — how probable this photometry is *if* the object is a quasar at z_0 — says nothing on its own about whether the object is a quasar. Turning it into a probability requires knowing how many quasars and how many other objects exist at this brightness. We compute and report both, and never let the first be read as the second.

Two hypotheses are not enough. A calculation that pits “quasar at z_0 ” against “star” assigns a high probability to a genuine quasar at $z = 2.6$ beside a primary at $z = 1.4$. For a pair search that is the dominant contaminant, so the competing hypotheses must include field quasars.

2 Hypotheses and intensities

Let $\mathbf{u}$ be the measured band luptitudes (§3), u_a the one chosen as reference, and $\mathbf{u}_{\setminus a}$ the others. We consider

H_{same}	a quasar whose redshift matches z_0 under a declared window
H_{field}	a quasar at any other redshift
H_{bkg}	anything else in the imaging, as described by the field model
H_{out}	a broad, fitted share of the field that neither model describes

H_{bkg} is deliberately not “star”. The population a chance projection draws from is everything the imaging contains, and it needs no selection function, because it is exactly the set of objects being counted (§5).

We work with *intensities*: expected counts per square degree, per unit reference luptitude, per unit volume of the other luptitudes. Writing $W(z \mid z_0) \in [0, 1]$ for the redshift-match weight, $\Sigma_Q(z, u_a)$ and $\Sigma_B(u_a)$ for the surface densities of objects measured in band a (§6), and $\eta(u_a)$ for the unmodelled share of the field (§8),

$$\lambda_{\text{same}} = \int W(z \mid z_0) \Sigma_Q(z, u_a) p(\mathbf{u}_{\setminus a} \mid u_a, Q, z) dz, \quad (1)$$

$$\lambda_{\text{field}} = \int [1 - W(z \mid z_0)] \Sigma_Q(z, u_a) p(\mathbf{u}_{\setminus a} \mid u_a, Q, z) dz, \quad (2)$$

$$\lambda_{\text{bkg}} = [1 - \eta(u_a)] \Sigma_B(u_a) p(\mathbf{u}_{\setminus a} \mid u_a, B), \quad (3)$$

$$\lambda_{\text{out}} = \eta(u_a) \Sigma_B(u_a) p(\mathbf{u}_{\setminus a} \mid u_a, U). \quad (4)$$

All four carry the same units, so their ratios are meaningful, and

$$p_{\text{same}} = \frac{\lambda_{\text{same}}}{\lambda_{\text{same}} + \lambda_{\text{field}} + \lambda_{\text{bkg}} + \lambda_{\text{out}}}. \quad (5)$$

Alongside this we always report the colour Bayes factor

$$\ln \text{BF} = \ln p(\mathbf{u}_{\setminus a} \mid u_a, Q, z_0) - \ln p(\mathbf{u}_{\setminus a} \mid u_a, \text{field}), \quad p(\cdot \mid \text{field}) = (1 - \eta) p(\cdot \mid B) + \eta p(\cdot \mid U), \quad (6)$$

which depends on the density models and on the field’s composition η but on neither surface density. If the surface densities change, (5) must move and (6) must not; a unit test enforces it.

3 Photometry

3.1 Forty-one bands, each in its own system

The model stores 41 individually labelled bands: SDSS *ugriz*; Legacy Surveys DR9 *g, r, z, W1, W2*, separately for the southern (DECam, releases 9010 and 9012) and northern (9011) systems; AllWISE *W1–W4*; PS1 *grizy*; NSC *ugrizy* and VR; SkyMapper *uvgriz*; and VHS *YJHK_s*. Filters from different surveys are never identified with one another. In particular the Legacy *W1, W2* — unWISE fluxes forced at the optical position and deblended with the optical model — and the AllWISE catalogue *W1, W2*, a separate detection with a $\sim 6''$ beam, are different measurements and are separate bands. The distinction matters: the Legacy forced *W1, W2* are the single strongest discriminant available (§10.3).

Each band keeps its native catalogue calibration. Optical magnitudes use their survey conventions; AllWISE and VHS retain Vega zero points; Legacy fluxes, including *W1, W2*, are AB nanomaggies. A native magnitude of 22.5 corresponds to flux one. AllWISE profile-fit DN fluxes use the documented zero points 20.5, 19.5, 18.0 and 13.0 for *W1–W4*. Magnitude-only tables supply measured magnitudes and errors; an absent magnitude is never converted into an invented flux or an upper limit. Photometry is *observed*, not dereddened: the training samples and the field sample are at $|b| \geq 25^\circ$, where the correction is small, and a candidate must be supplied the same way.

A band is usable only when its survey says so: finite flux and positive variance, a positive exposure count where the survey reports one, and the survey’s cleanliness flag (Legacy `maskbits`= 0, SDSS `clean`, NSC `flags`= 0, SkyMapper per-band `ngood` and `flags`, VHS per-band error bits, AllWISE `cc_flags`). An unusable band enters the observed-dimension mask and is marginalised exactly.

3.2 Luptitudes and their covariance

For native calibrated flux f_b we use the asinh magnitude

$$u_b = 22.5 - \frac{2.5}{\ln 10} \left[\operatorname{asinh} \left(\frac{f_b}{2s_b} \right) + \ln s_b \right], \quad (7)$$

with a softening s_b per band equal to the median flux error of the band in the training field sample, saved with the model. Luptitudes approach ordinary magnitudes at high signal-to-noise and stay finite for zero and negative flux. That matters: a faint source can have a negative measured flux, and that is a real measurement — faintness in WISE is among the strongest signals that something is *not* a quasar. Dropping such measurements, or replacing them with upper limits, would remove the most informative part of the competing population.

The covariance follows from the Jacobian $\partial u_b / \partial f_b = -(2.5 / \ln 10) / \sqrt{f_b^2 + 4s_b^2}$ applied to the full flux covariance, including any off-diagonal measurement correlations the caller supplies. This is first order; it is accurate while the flux error is small compared with $\sqrt{f_b^2 + 4s_b^2}$, which the softening guarantees at low flux.

3.3 Conditioning on a reference band

A density over all bands, brightness included, has the brightness distribution of the training sample inside it — a property of the target selection, not of quasar colour. Conditioning on one measured band removes it. Let O be the observed bands and $a \in O$ the first of them in a saved priority order (SDSS *r*, Legacy south *r*, Legacy north *r*, PS1 *r*, NSC *r*, SkyMapper *r*, VHS *J*, AllWISE *W1*, then the rest). For a quasar slice,

$$p(\mathbf{u}_{O \setminus a} \mid u_a, Q, z) = \frac{p(\mathbf{u}_O \mid Q, z)}{p(u_a \mid Q, z)}, \quad (8)$$

both obtained from the joint mixture after convolving the candidate’s measurement covariance, so the reference band’s own error is included and its correlation with the others is respected. Conditioning the whole mixture also re-weights its components; conditioning only the component means would be a different, incorrect density. At fixed u_a , replacing the other luminosities by the colours $u_b - u_a$ has unit Jacobian, so (8) is a colour density. Neighbouring redshift slices are interpolated in conditional density, which preserves normalisation. The field densities are conditioned on the same band.

The brightness itself enters only through the surface densities $\Sigma_Q(z, u_a)$ and $\Sigma_B(u_a)$, which is where it belongs: how many quasars and field sources exist at this brightness is a population statement, and the Bayes factor (6) stays free of it.

3.4 Missing bands

A Gaussian mixture marginalises exactly over unobserved dimensions: delete the corresponding rows and columns of every component covariance and of the observation’s covariance. No imputation is involved; the result is the likelihood of what was measured. Two measured bands supply one colour and can be scored when the caller asks for a minimum of two; a richer input is evaluated jointly, so separate survey Bayes factors are never multiplied under an independence assumption.

4 The quasar model

4.1 Training sample

The parent is DESI DR1 (`spectype='QS0'`, `zwarn=0`) and SDSS DR16Q (Lyke et al., 2020), de-duplicated by position at $1''$ — by position, not by identifier, because DESI DR1 alone repeats 9,577 quasars under two target identifiers — and restricted to $|b| \geq 25^\circ$: 1,856,961 quasars. From it we draw up to 2,000 per redshift interval of width 0.1 over $0.1 < z < 4.4$ (fewer in the four sparsest intervals), 82,489 in all, and match each to the seven catalogues within $1''$ ($2''$ for AllWISE). The draw is flat in redshift so that every slice is equally well constrained; within an interval it is uniform, so the brightness and colour distributions there are those of the parent. The sixteen inside-4 sky blocks reserved by the predecessor model are reserved again before anything is fitted, so the two can be compared on identical held-out objects; fifteen of them contain drawn quasars, 14,814 in all (14,766 with at least two usable bands). Component counts are chosen on sixteen further blocks inside the training sample, and 67,574 quasars are fitted.

DESI targets quasars with a classifier built on these very colours, so the training set describes $p(\mathbf{u} \mid Q, z, \text{selected})$. SDSS, whose targeting channels differ, supplies 18% of the drawn sample.

4.2 Extreme deconvolution

Each object is measured with its own heteroscedastic, correlated error. Fitting a mixture to the observed points would recover the locus *convolved* with those errors, which is systematically too wide, and the width is what discriminates. We fit the *intrinsic* distribution by extreme deconvolution (Bovy et al., 2011): a Gaussian mixture

$$p(\mathbf{v}) = \sum_k \alpha_k \mathcal{N}(\mathbf{v} \mid \boldsymbol{\mu}_k, \mathbf{V}_k) \quad (9)$$

whose components are convolved with each observation’s covariance $\mathbf{S}_i$ during fitting, so that the observed value is evaluated with $\mathbf{V}_k + \mathbf{S}_i$. The update is given in Appendix B. Figure 1 tests the principle on real quasar colours by injecting noise of known size: the deconvolved width is recovered to within 3% after the observed width has more than doubled, while an ordinary mixture tracks the broadened distribution exactly.

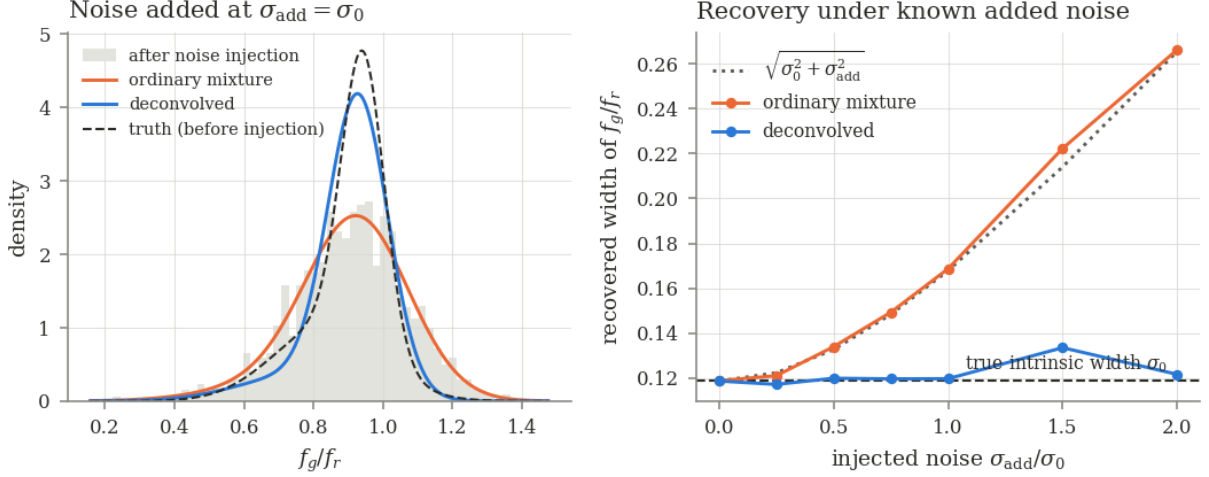


Figure 1: Deconvolution tested by controlled noise injection on real quasar colours (Legacy f_g/f_r of bright quasars near $z = 1.8$, $N = 1536$, $r < 20$, whose measurement errors are negligible; fitted width $\sigma_0 = 0.119$). Gaussian noise of known size is added and the models refitted. *Left:* at $\sigma_{\text{add}} = \sigma_0$ the deconvolved mixture (blue) recovers the pre-injection truth (dashed) while the ordinary mixture (orange) is broadened. *Right:* the ordinary mixture follows $\sqrt{\sigma_0^2 + \sigma_{\text{add}}^2}$, reaching 0.266 at $\sigma_{\text{add}} = 2\sigma_0$; extreme deconvolution returns 0.122. The test is of the algorithm, and was run in the predecessor’s flux-ratio coordinates (Appendix C).

Whose implementation. The algorithm is Bovy, Hogg & Roweis’s; the code is not. `qso_pcolor` carries its own numpy implementation, because we wanted marginalisation over an absent band to be exact and per object: a boolean mask selecting each object’s observed sub-block, where the reference implementation uses per-object projection matrices. With no missing dimensions the two agree by construction. Two defects were found and fixed during development — a covariance M step that cancelled catastrophically at flux scales of 10^8 , and a transposed cross term that corrupted off-diagonal covariances only when bands were missing — and both are pinned by regression tests against independent routes (quadrature, Monte Carlo, closed forms).

4.3 Redshift slices rather than one joint mixture

One could fit a single mixture to $[\mathbf{u}, z]$ and condition on z , as XDQSOz does (Bovy et al., 2012). Within a component the colour–redshift relation is then linear, so the strongly curved quasar track — emission lines and the Lyman break entering and leaving the passbands — needs components narrow in z , and conditioning on z re-weights components by their z -marginal, the redshift distribution of the training sample. We instead fit an independent deconvolved mixture in each of 43 overlapping slices, centred from $z = 0.15$ to 4.35 in steps of 0.1, each using quasars within half a slice width beyond its edges, and interpolate linearly in density:

$$p(\mathbf{u} \mid Q, z) = (1 - w) p_j(\mathbf{u}) + w p_{j+1}(\mathbf{u}), \quad w = \frac{z - z_j}{z_{j+1} - z_j}. \quad (10)$$

Each p_j is normalised, so (10) is a normalised density for any z , and the training sample’s redshift distribution *between* slices cannot leak into it; the redshift prior is supplied separately and visibly, as Σ_Q . Within a slice the fit is still an average over the training redshifts there. Component counts are chosen per slice by held-out density on the selection blocks from $K \in \{1, 2, 4\}$: $K = 4$ in 33 slices and $K = 2$ in 10. Outside $0.15 \leq z \leq 4.35$ the model has no slice; the scorer integrates only inside that support and reports, as `frac_norm_outside_support`, how much of the redshift normalisation a wider grid would have taken from outside it.

The fits stop at a 150-iteration cap; one of 43 reaches the 10^{-5} tolerance. The validation below evaluates the saved fits at that stopping point.

Figures 2 and 3 show ten one-colour marginals of the fitted model over its whole support.

For a Gaussian component with mean μ and covariance $\mathbf{V}$, the exact marginal of a colour $u_a - u_b$ has mean $\mu_a - \mu_b$ and variance $V_{aa} + V_{bb} - 2V_{ab}$; the figures use this with the fitted weights and the slice interpolation, without smoothing. Numerical integration of the displayed grids agrees with the analytic mixture CDF to 2×10^{-7} , and the axis limits contain at least 99.5% of the probability at every redshift.

5 The field model

5.1 Everything a chance neighbour can be

The competing density is built from the surveys themselves. We take 24 cones of radius 0.15° at $|b| \geq 25^\circ$, spread in declination and over distinct sky blocks, query all seven catalogues in each, and form the positional union: sources are associated across catalogues within $1''$ ($2''$ for AllWISE), one-to-one and nearest first, and an unmatched source is kept as a source of its own. No morphological or star/galaxy selection is applied, because the candidates have none: any selection applied to the field sample that the candidates do not share is a bias in the Bayes factor. The predecessor model measured the cost of getting this wrong: a background built from point sources alone made galaxies look like a serious contaminant (quasar/galaxy AUC 0.80), while an all-source background gave 0.96 (Appendix C).

Known quasars are removed, and it matters more than it looks. H_{field} and H_{bkg} must be mutually exclusive. Quasars are a tiny fraction of all sources, but what matters is their fraction *in quasar-like colour space*, where the same-redshift hypothesis is tested. Measured on a one-degree Legacy cone with the predecessor model:

ln BF (quasar at z_0 / background)	sources	known quasars
−40 to −10	17,995	0.1%
−5 to −2	6,026	0.5%
0 to +2	978	9.5%
+2 to +5	725	31.2%
+5 to +40	272	72.4%

Left in, the field model learns the quasar locus and competes against the quasar hypothesis using quasars, costing about 1.3 in ln BF. DESI DR1 and SDSS DR16Q quasars within $1''$ are therefore removed from every field cone. Unobserved quasars remain, a residual of the same bias in the same direction; they are meant to be carried by Σ_Q through a completeness (§6).

After quality cuts and quasar removal the fields hold 174,567 sources with at least two usable bands; six whole fields (42,493 sources) are reserved for validation, four more choose the number of components, and a further reserved overlap field (6,825 sources) supplies survey combinations absent from the six.

5.2 Rare measurement patterns

Which bands a field source has depends on where it is and how bright it is, and some combinations are rare. The fit uses 20,000 training sources drawn so that every observed-band pattern is represented by at least 20 sources where it has them, with inverse sampling weights restoring each pattern’s abundance in the full field sample. This keeps rare combinations in the fit without promoting them in the field distribution. Component selection over $K \in \{4, 8, 12\}$ chooses $K = 8$; the fit stops at the 150-iteration cap.

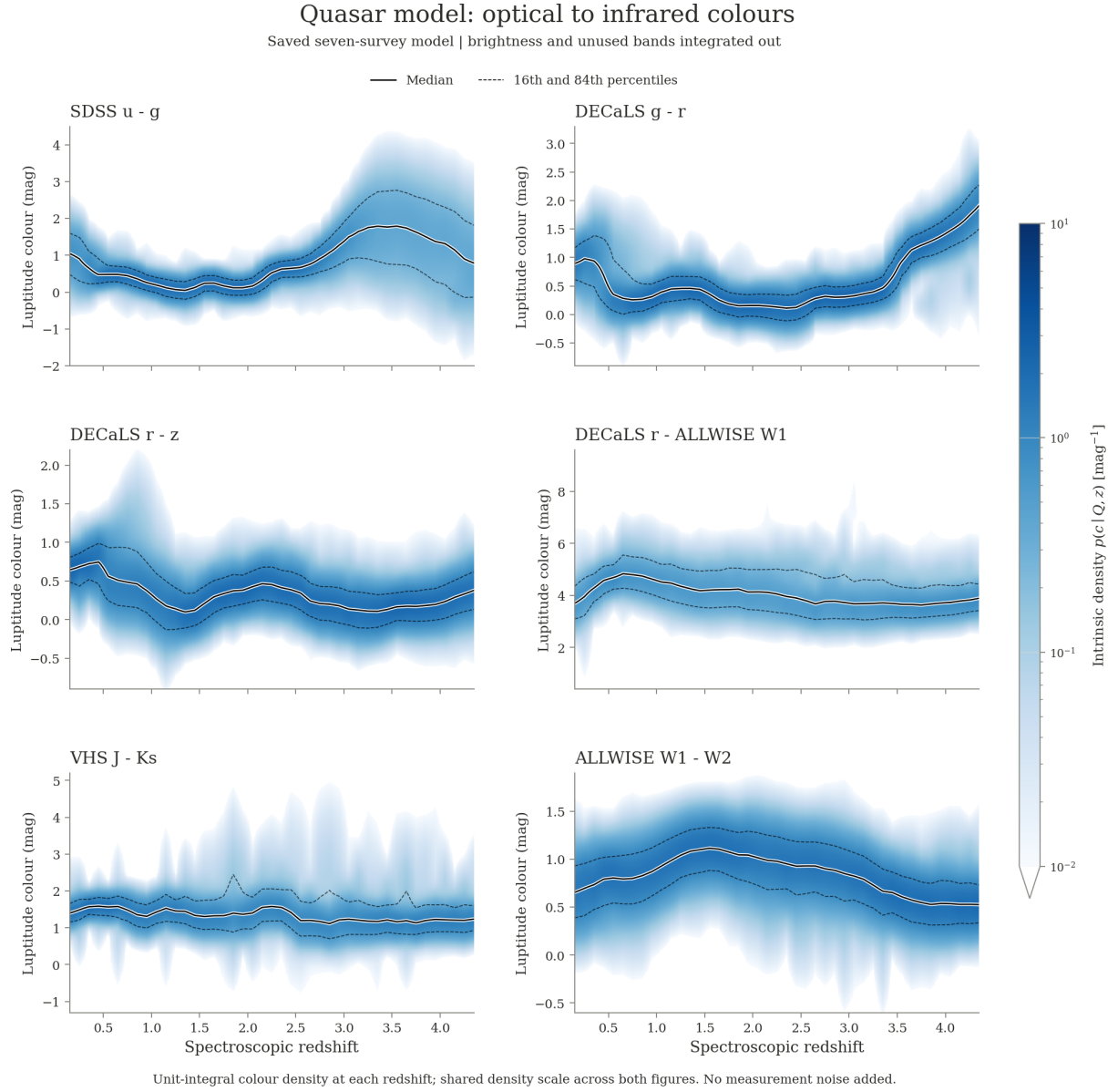


Figure 2: Intrinsic quasar colour densities from the model: SDSS $u - g$, Legacy $g - r$ and $r - z$, Legacy $r - \text{AllWISE } W1$, VHS $J - K_s$ and AllWISE $W1 - W2$. Brightness and all unused bands are marginalised; no measurement noise is added. Each redshift column is a unit-integral colour density, per magnitude; solid curves are medians and dashed curves the 16th and 84th percentiles. The density scale is shared with Figure 3.

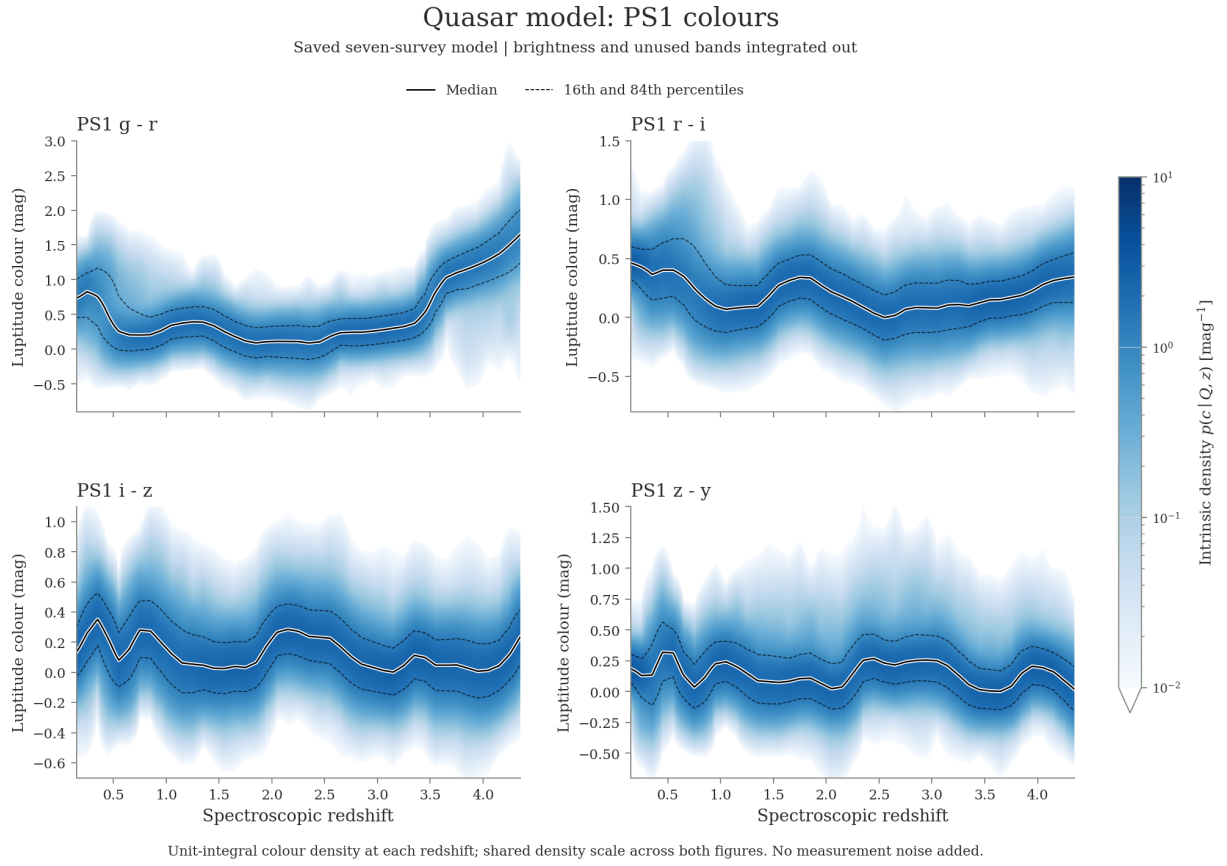


Figure 3: The four adjacent PS1 colours, $g - r$, $r - i$, $i - z$ and $z - y$, against redshift, drawn as in Figure 2. These use the separately labelled PS1 bands; they are not Legacy colours relabelled.

5.3 Dedicated fits for Legacy-only input

A single 41-band mixture of eight components is a coarse description of a field population that varies a great deal between surveys’ depths, and it under-represents the stellar locus when only Legacy bands are supplied. The model therefore also stores separately fitted joint densities over two band sets: Legacy south g, r, z and Legacy south $g, r, z, W1, W2$, each fitted to the same weighted training rows (13,239 sources with all of its bands, 2,597 of them in the selection fields), with its component count chosen on those fields from $K \in \{8, 16, 32\}$: both choose $K = 32$, by 0.001 and 0.016 nats per object over $K = 16$, and both converge (in 60 and 84 iterations). A dedicated fit is used only when *every* observed band lies in its band set, the smallest covering set first; otherwise the 41-band background is used. Routing depends on which bands were measured, never on their values or on class. Each dedicated fit is a normalised density conditioned on the reference band exactly as (8), and is an alternative fitted approximation, not the projection of the joint one. The quasar density always uses the joint quasar mixtures.

The effect is measured on the labelled pairs of §10. With g, r, z alone, the dedicated fit raises the held-out quasar/star AUC from 0.795 under the 41-band background to 0.874. With $g, r, z, W1, W2$ it moves quasar/star from 0.981 to 0.982 and quasar/galaxy from 0.956 to 0.961, and lowers the same-/wrong-redshift AUC by $\ln R$ from 0.833 to 0.828: a better field density enters the denominator of R for quasars too.

6 Surface densities, per reference band

$\Sigma_Q(z, u_a)$ and $\Sigma_B(u_a)$ must be densities per unit native luptitude of the reference band, for objects *with that band measured*, because that is the condition under which the conditional densities of §3.3 are evaluated. They are built separately for each of the 41 bands (`scripts/build_multisurvey_priors.py`); 37 have enough data, and a candidate whose reference is one of the other four (NSC u , SkyMapper u, v , VHS Y) receives the no-prior status, with evidence but no posterior.

Footprints. A band can be missing because the object is outside the survey or because it is too faint. Only the second is information about the population, so both densities are counted inside the footprint of the survey that supplies the reference band. For the field cones the footprint is unambiguous: a survey’s detected fraction in a cone is either zero or not (measured). For quasars, a cell of HEALPix inside 32 (3.4 deg^2) is in the footprint if the survey detected at least two quasars in it; shallow surveys such as SkyMapper and VHS miss many faint quasars even inside their footprint, which is why presence rather than a detected fraction defines it.

Quasars. Each quasar of the flat draw is weighted by $N_{\text{parent}}(\text{bin})/n_{\text{drawn}}(\text{bin})$, which restores the parent’s redshift distribution exactly (the weights sum to the 1,856,961 parent quasars); within a redshift bin the draw is uniform, so the brightness distribution is the parent’s. The weighted counts in (redshift 0.1, luptitude 0.5) cells, inside the footprint, are divided by the footprint’s share of the parent coverage ($13,825 \text{ deg}^2$ in all; 10,048 for Legacy south):

$$\Sigma_Q(z, u_a) = C \frac{\sum_i w_i [z_i, u_{a,i} \in \text{cell}, a \text{ measured}, i \in \text{footprint}_a]}{A_a \Delta z \Delta u}. \quad (11)$$

One completeness constant. The parent is spectroscopically incomplete, unevenly over the sky. We apply one constant C to every band, fixed so that the Legacy-south r density integrates over $17 \leq u_r < 22.5$ to the predecessor’s validated prior, which was normalised to DESI’s best-covered sky: $C = 2.30$. That assumes the completeness does not depend on which band is the reference.

Field. $\Sigma_B(u_a)$ is the number of clean field sources with band a measured, per unit u_a , divided by the area of the cones that survey covers.

Checks. The Legacy-south values reproduce the predecessor’s on different data. Its independently counted field density over $17 \leq r < 22.5$ is $18,945 \text{ deg}^{-2}$, against 20,661 from eight 0.5° Legacy cones with dereddened photometry (-8% , of the expected sign: dereddening brightens sources past the limit). Quasar densities seen through different surveys’ r bands agree without being forced to:

reference band	$\Sigma_Q, 17 \leq u_a < 22.5 \text{ (deg}^{-2}\text{)}$	$\Sigma_B, \text{ same range (deg}^{-2}\text{)}$
Legacy south r	254.9	18,945
Legacy north r	212.1	13,415
SDSS r	245.1	12,312
PS1 r	263.0	15,021
NSC r	216.1	12,577

The field densities differ, as they should: Σ_B counts sources *measured* in that band, and the shallower surveys measure fewer. A unit test requires the quasar densities over $17 \leq u_a < 21$ to agree within 20%.

What a completeness does and does not change. Multiplying Σ_Q by a constant leaves $p(z \in W \mid \mathbf{u}, Q)$ unchanged, because both quasar terms scale together, but not the competition with the field: with total quasar intensity Q and field intensity B , p_{same} and R change by $a(Q + B)/(aQ + B)$, which depends on the candidate. A redshift-*dependent* completeness changes the balance between λ_{same} and λ_{field} as well: intensities $(1, 1, 1)$ give $p_{\text{same}} = 1/3$, while a correction multiplying only the field term by 100 gives $1/102$. C is a declared, reproducible choice, not a conservative one. Where no prior pair exists for the reference band, the scorer returns the evidence and leaves the posterior fields null with a status code, rather than substituting a class fraction.

7 The redshift window is a multiplicative constant

This section contains the main methodological point.

7.1 The tension

A physically motivated pair window is of order $\pm 2000 \text{ km s}^{-1}$; at $z_0 = 1.4$

$$\Delta z = (1 + z_0) \frac{\Delta v}{c} = 2.4 \times \frac{2000}{299792.458} = 0.016. \quad (12)$$

Broadband photometry constrains a quasar redshift far more loosely, and often to a multimodal distribution — which is why XDQSOz produces a redshift *PDF* (Bovy et al., 2012). With Legacy $g, r, z, W1, W2$ the median posterior width was $\sigma_z \approx 0.6$ (measured on 1500 DESI quasars with the predecessor model). At $z_0 = 1.8$ the window spans 0.037 in redshift; for a unimodal posterior of width 0.6 it captures of order two per cent of the probability (Figure 4). So $\lambda_{\text{field}} \gg \lambda_{\text{same}}$ and p_{same} is small even for an object exactly on the locus at exactly the right redshift. That is the correct answer to the question as posed, and it means the question as posed is not the one colours can answer.

7.2 Which uncertainty is the problem

	$\Delta z \text{ at } z = 1.8$	relative
primary’s spectroscopic error	0.0004	$1 \times$
$\pm 2000 \text{ km s}^{-1}$ window (half-width)	0.019	$49 \times$
companion’s photometric width	0.6	$1579 \times$

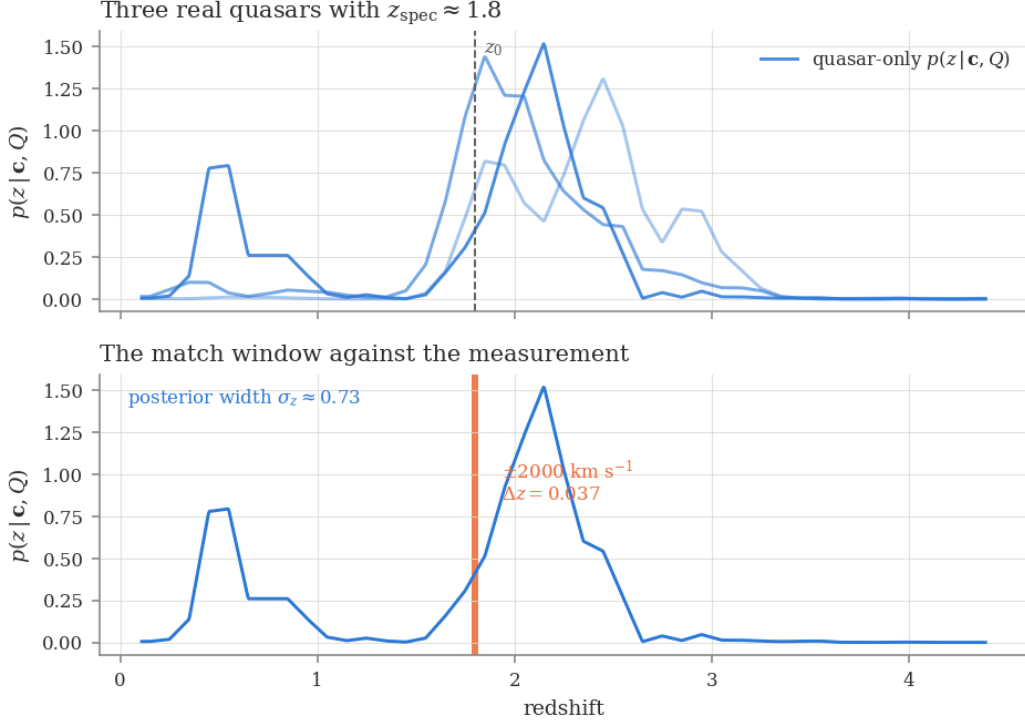


Figure 4: *Top*: quasar-only redshift posteriors for three DESI quasars with $z_{\text{spec}} \approx 1.8$ under a flat prior, showing the width and the secondary solutions broadband colours cannot resolve. *Bottom*: the same posterior with a $\pm 2000 \text{ km s}^{-1}$ window drawn to scale (narrow orange band). Computed with the predecessor Legacy model; the property shown — a window far narrower than the posterior — is one of the bands, not of the model.

The median DESI DR1 quasar redshift error is 0.0004 (43 km s^{-1}). The primary is known some 1600 times better than the companion; the difficulty sits entirely on the companion. What the primary’s precision buys is that $p(\mathbf{u} | Q, z_0)$ is evaluated at a sharp point. Its error enters by *marginalising* the window over it, which smooths the window’s edges while conserving its area; an earlier implementation replaced the half-width by $\sqrt{s^2 + \sigma_{\text{primary}}^2}$, which increased the area, so a less certain primary produced more same-redshift companions. The 43 km s^{-1} is a pipeline statistical error; quasar systemic redshifts from broad lines, C IV in particular, carry systematic offsets that are a non-trivial fraction of a pair window, and `z_primary_err` should be set to a systemic uncertainty measured for the estimator actually used.

7.3 What sets the companion’s width

To first order

$$\sigma_z \approx \frac{\text{scatter of colour at fixed } z}{|d(\text{colour})/dz|}. \quad (13)$$

Near $z = 1.8$, Legacy $g - r$ moves at 0.356 per unit redshift with scatter 0.219 , and $r - z$ at 0.402 with 0.229 : implied widths 0.62 and 0.57 . At bright magnitudes the scatter is intrinsic quasar-to-quasar diversity, not measurement error, so deeper imaging in the same bands buys little. Fifteen times more training data left σ_z unchanged to 0.001 (Appendix C). Improving the redshift constraint needs bands in which quasar colour changes *faster* with redshift: a u band spanning the Lyman break, or a narrow band on a strong line. This is part of the case for combining surveys, and Table 3 shows that combinations including SDSS u discriminate redshift better.

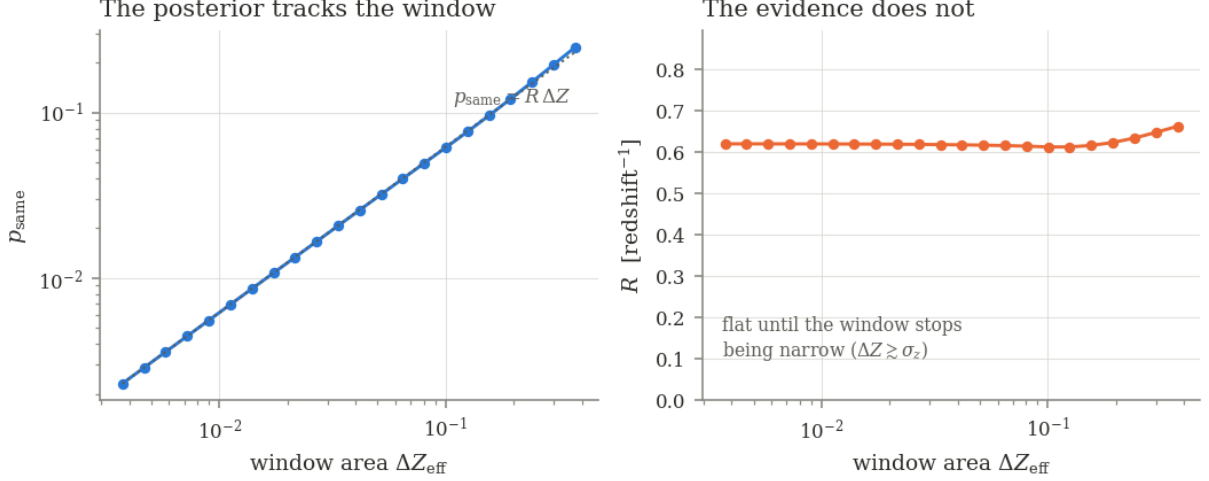


Figure 5: The factorisation on a real quasar. *Left*: p_{same} is proportional to the window area over more than a decade, as (17) requires; the dotted line is $R \Delta Z$ with R from the narrowest window. *Right*: R is flat until the window stops being narrow compared with σ_z . Computed with the predecessor model; the factorisation is a property of (1)–(15), not of a particular density.

7.4 Factorising the window out

Because the window is much narrower than the scale on which $\Sigma_Q p(\mathbf{u} | Q, z)$ varies, (1) becomes

$$\lambda_{\text{same}} \simeq \Delta Z_{\text{eff}} \Sigma_Q(z_0, u_a) p(\mathbf{u}_{\setminus a} | u_a, Q, z_0), \quad \Delta Z_{\text{eff}} \equiv \int W(z | z_0) dz. \quad (14)$$

The step that makes this clean is the denominator. H_{same} and H_{field} *partition* the quasar intensity, so

$$\lambda_{\text{same}} + \lambda_{\text{field}} + \lambda_{\text{bkg}} + \lambda_{\text{out}} = \Lambda_{Q,\text{tot}} + \lambda_{\text{bkg}} + \lambda_{\text{out}} \quad \text{for any window,} \quad (15)$$

with $\Lambda_{Q,\text{tot}} = \int \Sigma_Q p(\mathbf{u}_{\setminus a} | u_a, Q, z) dz$. Defining

$$R \equiv \frac{\Sigma_Q(z_0, u_a) p(\mathbf{u}_{\setminus a} | u_a, Q, z_0)}{\Lambda_{Q,\text{tot}} + \lambda_{\text{bkg}} + \lambda_{\text{out}}} \quad [\text{redshift}^{-1}] \quad (16)$$

gives

$$p_{\text{same}} = R \Delta Z_{\text{eff}}, \quad (17)$$

a linear relation, not an odds transformation.

Two cautions. First, (15) is exact but (14) is an approximation: writing $f(z) = \Sigma_Q(z, u_a) p(\mathbf{u}_{\setminus a} | u_a, Q, z)$,

$$p_{\text{same}} = R(z_0) \Delta Z_{\text{eff}} + \frac{\int W(z) [f(z) - f(z_0)] dz}{D}, \quad D = \int f dz + \lambda_{\text{bkg}} + \lambda_{\text{out}}, \quad (18)$$

so the linear relation holds to the extent that f is flat across the window. The leading relative correction is $f''(z_0)s^2/[6f(z_0)]$ for a top hat of half-width s ; at a cusp of the piecewise-linear interpolation it is $O(s)$. Second, the implementation stores the *window-averaged* $R = p_{\text{same}}/\Delta Z_{\text{eff}}$, so (17) holds for the reported numbers by construction. The substantive claim is that R does not depend on the window, which holds in the narrow limit: over a factor of ten in window width R moves by well under a per cent (Figure 5).

7.5 Consequences

1. **Rank on R .** It is the window-free evidence, so two people can compare candidates without agreeing on a window, and it is not compressed towards zero the way p_{same} is.
2. **Never quote p_{same} without its window.** Every output row carries both, as `p_sameq` and `dz_match_eff`.
3. **Never integrate a velocity window on the main redshift grid.** $\Delta Z_{\text{eff}} = 0.03$ on a grid of step 0.01 is two or three points. The scorer inserts dense window nodes, exact support boundaries and the model and prior interpolation knots into one quadrature grid shared by the window and the total quasar intensity, so a prior peak resolved by one is resolved by both, and nothing is integrated outside the common support.
4. **Widening the window does not improve the answer.** It changes the question.

8 Colour outliers: the unmodelled hypothesis

8.1 The defect

Both colour densities are Gaussian mixtures convolved with the object’s covariance $\mathbf{S}$. Far from every component of both, each is dominated by its widest component in the direction of the displacement $\Delta \mathbf{u}$, and their log ratio becomes a difference of quadratic forms,

$$\ln \frac{p(\mathbf{u} \mid Q, z)}{p(\mathbf{u} \mid B)} \approx -\frac{1}{2} \Delta \mathbf{u}^T \left[(\mathbf{V}_Q + \mathbf{S})^{-1} - (\mathbf{V}_B + \mathbf{S})^{-1} \right] \Delta \mathbf{u} + \text{const}, \quad (19)$$

which grows without bound in any direction where the widest quasar component is wider than the widest field component. Neither model was fitted to objects there, so (19) reflects how two fits’ tails fell, not evidence. With three hypotheses an artefact, a moving object or a population absent from both training sets is called a quasar with probability one.

This was found and measured on the predecessor model. A real candidate ($z_0 = 1.8$, $r \approx 21$) with its g flux multiplied by 8 at fixed error lay 6.6σ from every quasar component and 14.8σ from every field component, and was a quasar with probability 1.000, $\ln \text{BF} = +24.7$; multiplied by 32, 22σ from every quasar component, likewise. Among the 21,284 spectroscopic non-quasars of the pair validation, 11 lay more than 3σ from both models, and 7 of those were called quasars, against 0.9% near the loci.

Two distances identify the regime and are reported for every object: the Mahalanobis distance to the nearest component,

$$d = \min_k \sqrt{(\mathbf{u} - \boldsymbol{\mu}_k)^T (\mathbf{V}_k + \mathbf{S})^{-1} (\mathbf{u} - \boldsymbol{\mu}_k)}, \quad (20)$$

conditioned on the reference band, over every component of every quasar slice (d_Q , `qso_ood_sigma_any_z`) and over the field density actually used (d_B , `bkg_ood_sigma`). The distance to the z_0 slice alone, `qso_ood_sigma`, is large for every wrong-redshift quasar and cannot identify this regime. $\min(d_Q, d_B)$ large means every density in the calculation is an extrapolation; the flag `outside_both_models` is raised above a threshold the caller must state.

8.2 A fourth hypothesis

The field is written as a modelled part and a broad unmodelled part U — the standard device for data containing points no component describes (Hogg et al., 2010, §3):

$$p(\mathbf{u} \mid \text{field}) = (1 - \eta) p(\mathbf{u} \mid B) + \eta p(\mathbf{u} \mid U), \quad p(\mathbf{u} \mid U) = \mathcal{N}(\mathbf{u} \mid \bar{\mathbf{u}}, \kappa^2 \mathbf{E} + \mathbf{S}), \quad (21)$$

one joint Gaussian over all 41 bands, conditioned on the reference band like everything else. It gives λ_{out} of (4), which joins every denominator, and the Bayes factor (6) compares against the whole field. With $\eta = 0$ the scorer reduces exactly to three hypotheses (a unit test).

The shape must dominate every component. If $\kappa^2 \mathbf{E} - \mathbf{V}_k \succ 0$ for every component k — every quasar slice, the joint field fit, and each dedicated Legacy fit embedded with zeros in the other bands — then U eventually dominates every component in every direction, for any noise and in every observed subspace. Conditioning preserves this: Schur complements are monotone in the Loewner order, because $(\mathbf{A}/\mathbf{A}_{aa})^{-1} = (\mathbf{A}^{-1})_{a \setminus a}$ and $\mathbf{A} \succeq \mathbf{B} \succ 0$ implies $\mathbf{A}^{-1} \preceq \mathbf{B}^{-1}$ (tested). The smallest such κ is $\kappa_{\min}^2 = \max_k \lambda_{\max}(\mathbf{V}_k, \mathbf{E})$, from a Cholesky factor of $\mathbf{E}$.

A scaled field covariance would not do: in the predecessor’s flux-ratio coordinates a few rare, infrared-bright quasar components were 67 times wider than the field, and an isotropic scaling would have made U 67 times too broad in every other direction. $\mathbf{E}$ is therefore an anisotropic envelope. Whiten every component by the field covariance $\Sigma_B = \mathbf{L}\mathbf{L}^T$, $\mathbf{A}_k = \mathbf{L}^{-1}\mathbf{V}_k\mathbf{L}^{-T}$; rotate to the eigenbasis $\mathbf{Q}$ of the mean $\mathbf{A}_k$; and set

$$\mathbf{E} = \mathbf{L}\mathbf{Q} \text{diag}(D) \mathbf{Q}^T \mathbf{L}^T, \quad D_i = \max\left(1, \max_k \sum_j |(\mathbf{Q}^T \mathbf{A}_k \mathbf{Q})_{ij}|\right). \quad (22)$$

A diagonal matrix of absolute row sums exceeds the symmetric matrix they come from (the difference is diagonally dominant with a non-negative diagonal), so $\mathbf{E} \succeq \mathbf{V}_k$ for every k and $\mathbf{E} \succeq \Sigma_B$. For the shipped model $\kappa_{\min} = 0.861$ over 224 components: 43 slices of two or four, eight in the joint field fit, and 32 + 32 in the Legacy fits.

8.3 Calibration

κ and η are set by field likelihood (`scripts/fit_multisurvey_outlier.py`). The fitting set (A) is the 112,074 training-field sources that the field fit did not use — it drew 20,000 — and the check set (B) is the 42,493 sources of the reserved fields. For each κ , η is fitted by EM in each of three reference-luptitude bins, separately for each reference band with enough objects, and once over all objects as the fallback for the rest; the log-likelihood is concave in η , so EM reaches the global maximum (checked against a direct optimiser in a unit test).

$\kappa/\kappa_{\min}$	1.02	1.25	1.5	2	3	5
gain on A (nats per object)	+0.025	+0.024	+0.022	+0.021	+0.019	+0.017
gain on B (nats per object)	+0.077	+0.072	+0.069	+0.064	+0.058	+0.051

The gain is the mean per-object improvement over the unmodified field density, $\ln[(1-\eta)p_B + \eta p_U] - \ln p_B$. The likelihood prefers the narrowest U that still dominates, so the dominance constraint binds and $\kappa = 1.02 \kappa_{\min}$ is adopted. The gain on the reserved fields is three times that on A: the 41-band field fit describes fields it was not fitted to less well, and U absorbs part of that. Six reference bands have their own η , in three luptitude bins each: SDSS r $(1.3, 0.5, 0.2) \times 10^{-3}$, Legacy south r $(3.7, 0.7, 1.2) \times 10^{-3}$, Legacy north r $(2.4, 0.2, 0.0) \times 10^{-3}$, AllWISE $W1$ $(0.0, 0.9, 0.0) \times 10^{-3}$, PS1 g $(34, 0, 1.1) \times 10^{-3}$ and PS1 r $(29, 3.3, 4.3) \times 10^{-3}$; the rest use the pooled 2.3×10^{-3} . The large PS1 values are in the brightest bin, where the field fit is least constrained.

8.4 What it changes

Far from both models it decides. In the sweep (Table 1, Figure 6) every step beyond about 4σ from the nearer model goes to U , and the Bayes factor turns strongly negative. Among the 95 spectroscopic non-quasars more than 3σ from both models, 17 are called quasars with three hypotheses and none with U ; between 2 and 3σ , 14 of 1,060 become 8. In total the non-quasars called quasars fall from 191 to 168 of 21,284.

Near the loci it changes little. On the 50,746 companions the median $|\Delta \ln R|$ is 0.003 and 99.9% move by less than 0.64; 105 move by more than 0.1. Every held-out AUC is unchanged to three decimals except same-/wrong-redshift by $\ln \text{BF}$, which rises from 0.714 to 0.720. U is more

band	s	d_Q, d_B	three hypotheses			with U		
			$\ln \text{BF}$	$P(Q)$	$\ln R$	$\ln \text{BF}$	$P(Q)$	$P(U)$
–	1	0.4, 2.3	+3.9	0.58	−1.73	+3.9	0.58	0.000
g	4	3.5, 5.3	+14.0	1.000	−2.5	+3.0	0.67	0.33
g	8	4.8, 7.3	+14.4	1.000	−8.3	−11.1	0.001	0.999
g	32	7.5, 11.2	+14.4	1.000	−36.8	−57.3	0.000	1.000
z	16	3.8, 6.6	−30.9	1.000	−60.6	−60.5	0.009	0.991
$W2$	8	2.4, 3.6	+11.2	0.999	−0.18	+5.7	0.76	0.24
$W2$	32	2.9, 4.7	+13.4	1.000	−8.5	−5.1	0.19	0.81

Table 1: The README candidate (Legacy south $g, r, z, W1, W2$, $z_0 = 1.8$, $r \approx 21$) with one band flux multiplied by s at fixed error. $P(Q)$ is the posterior probability of a quasar at any redshift; d_Q and d_B are in σ . With three hypotheses the candidate is a certain quasar however far it is pushed, and $\ln \text{BF}$ saturates at +14; with U it goes to “unmodelled”. The full sweep is in `data/outlier_sweep_multisurvey.json`.

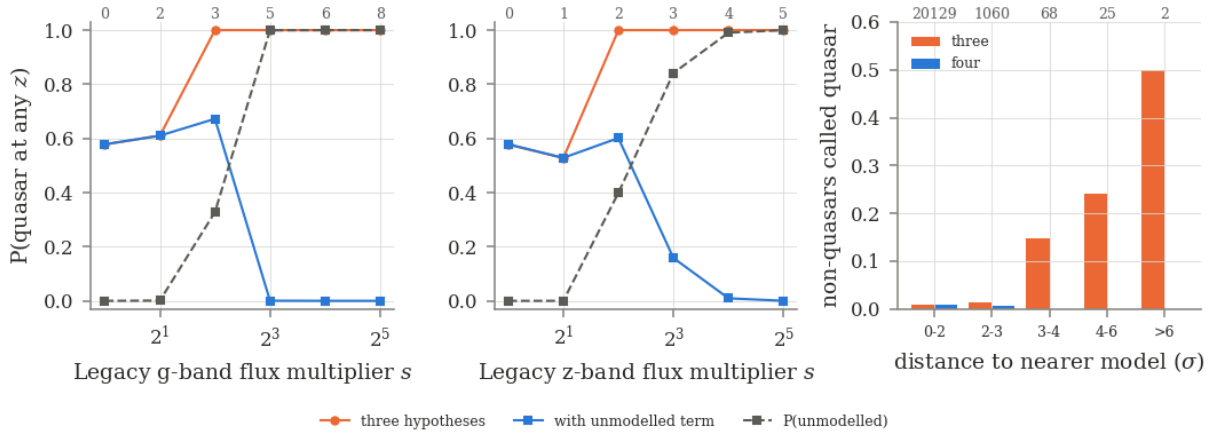


Figure 6: *Left, centre:* the README candidate with its Legacy g or z flux multiplied by s . Orange: three-hypothesis probability of a quasar at any redshift; blue: the same with U ; grey dashed: $P(U)$. Numbers along the top are $\min(d_Q, d_B)$ in σ . *Right:* the 21,284 spectroscopic non-quasars of the pair validation, in bins of distance to the nearer model (counts along the top): the fraction called quasars. With U it is zero in every bin beyond 3σ .

active here than in the predecessor’s four flux ratios: a conditional 41-band density leaves more room for the field fit to be poor, and the η fitted on held-out fields says so.

What it costs. Real quasars far from the field model pay: of the 15 spectroscopic quasars more than 3σ from both models, 7 are called quasars with three hypotheses and 2 with U , and between 2 and 3σ , 50 of 94 become 42. Those colours are outside what either model was fitted to, and the scorer declines to treat them as evidence. η belongs to this field model; a refitted or local one needs its own.

9 What the scorer returns

Every candidate produces a record separating evidence from posterior, with a status code and quality flags rather than a silent number.

<code>loglike_qso_zprimary</code>	$\ln p(\mathbf{u}_{\setminus a} \mid u_a, Q, z_0)$; a likelihood, not a probability
<code>loglike_bkg</code>	the field density of (6)
<code>log_bayes_factor_qz_bkg</code>	(6); independent of both surface densities
<code>p_zmatch_given_qso</code>	$P(z \in W \mid \mathbf{u}, Q)$ under Σ_Q (flat without a prior); conditional on being a quasar, normalised over the trained range
<code>z_phot_mode</code>	mode of the quasar-only redshift posterior
<code>frac_norm_outside_support</code>	share of that normalisation a wider grid would have taken from outside the trained range; it is <i>discarded</i>
<code>log_r_per_unit_z</code>	$\ln R$ of (16); the ranking statistic
<code>dz_match_eff</code>	ΔZ_{eff} actually used, so that $p_{\text{same}} = R \Delta Z_{\text{eff}}$
<code>log_lambda_sameq, _fieldq, _bkg, _out</code>	(1)–(4), as natural logs
<code>p_sameq</code>	(5)
<code>p_sameq_vs_bkg</code>	same-redshift quasar against every non-quasar explanation; ignores field quasars
<code>p_outlier, outlier_fraction, loglike_outlier</code>	the unmodelled share of the posterior, η , and $\ln p(\mathbf{u}_{\setminus a} \mid u_a, U)$
<code>qso_ood_sigma</code>	d of (20) to the z_0 slice
<code>qso_ood_sigma_any_z, bkg_ood_sigma</code>	d_Q and d_B
<code>reference_band, bands_used, surveys_used</code>	what was measured and what the densities were conditioned on
<code>n_bands_used, status, quality_flags</code>	why a value is missing, and what is suspect about one that is not
<code>identification</code>	<code>candidate_id, primary_id, ref_mag, photometric_system, model_manifest_id</code>

Status values:

- `ok, insufficient_photometry, blended_not_scored`;
- `no_prior_posterior_unavailable, qso_prior_empty_at_this_magnitude`;
- `primary_z_outside_model_support, window_outside_model_support`.

10 Validation

Nothing above is evidence that the method works. That requires labelled objects, and two kinds are used: pairs in which both members have spectra, and reserved quasars and field sources for every survey combination.

10.1 Spectroscopic pairs: design

Around each DESI DR1 quasar we take every companion at 3–30'' that has a DR1 spectrum of its own and label it by what that spectrum says: a quasar within $\pm 3000 \text{ km s}^{-1}$ of the primary (`same_z`), a quasar outside (`field_q`), or anything else (`non_qso`). The labels are spectroscopic and so independent of the photometry being scored. The *inclusion* of an object is not: DESI targets quasars with a classifier built on these colours, so the labelled set is a colour-selected subset of the sky, and its calibration transfers only to candidates drawn the same way. The non-quasars in particular are DESI targets, not a random draw of the imaging.

From 1,645,842 DESI DR1 quasar primaries, 579,572 companions have a usable spectrum. We require Legacy south photometry, a primary inside the trained redshift support, `maskbits`=0, reference-band `fracflux_r` ≤ 0.2 , $17 \leq r < 22.5$ and usable g, r, z ; the non-quasars are subsampled to 30,000 with seed zero. That leaves 50,746 companions: 3,308 `same_z`, 26,154 `field_q`, 21,284 `non_qso`. The 14,221 in the sixteen reserved sky blocks — 880, 7,293 and 6,048 — are held out from the quasar fit; ranking AUCs use them. Every companion is scored from its Legacy $g, r, z, W1, W2$ fluxes as observed (`scripts/validate_pairs_multisurvey.py`).

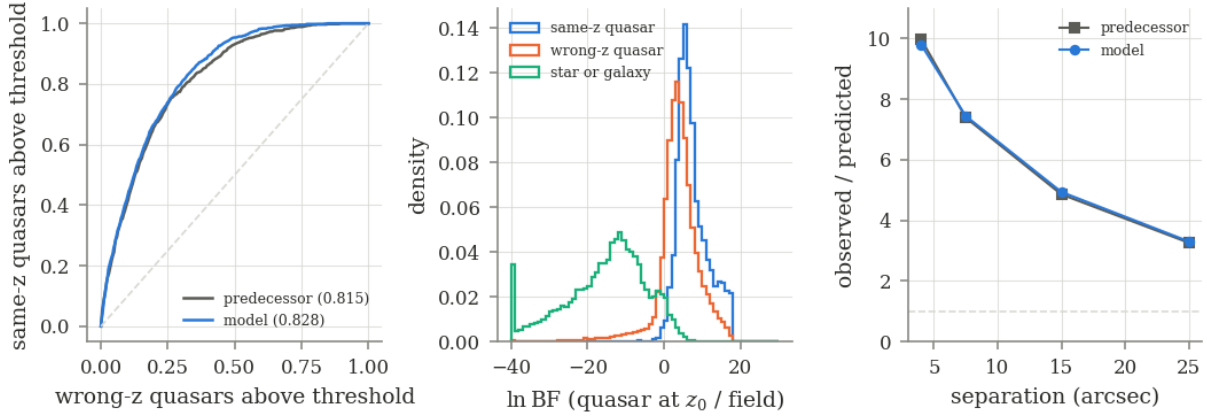


Figure 7: The labelled pairs, scored from Legacy $g, r, z, W1, W2$. *Left*: ROC for same- against wrong-redshift quasar companions in the reserved sky blocks, ranked by $\ln R$, for the model and its predecessor (AUC in brackets). *Centre*: the Bayes factor for the three classes (values below -40 are piled at -40); the non-quasar class mixes stars with the galaxies DESI targeted. *Right*: the observed same-redshift fraction among quasar companions divided by the mean predicted $p(z \in W \mid \mathbf{u}, Q)$, by separation. A defect in the scorer would give a flat ratio; physical clustering gives one that falls with separation.

10.2 Spectroscopic pairs: results

Redshift discrimination. Ranking the held-out quasar companions by $\ln R$, the AUC for `same_z` against `field_q` is **0.828**; by $p(z \in W \mid \mathbf{u}, Q)$ alone — the quasar-only ranking, which no field density enters — it is 0.865; by the Bayes factor, which has no field-quasar term, 0.720 (Figure 7). This is discrimination against the whole wrong-redshift population, not velocity resolution.

Quasars just outside the window. Against the 151 held-out quasars with $3000 \leq |\Delta v| < 6000 \text{ km s}^{-1}$, the 880 same-redshift quasars give AUC 0.472 by $\ln R$ — chance. The photometry does not resolve the match boundary. This is a limit of the information, not something refitting would change.

The quasar finder. Against spectroscopic stars the Bayes factor has AUC **0.982**, and 0.04% of stars exceed the median same-redshift quasar; against spectroscopic galaxies, **0.961** and 0.18%. Against all non-quasars, 0.966; ranking by $\ln R$ instead, same-redshift quasars against non-quasars, 0.995.

Calibration, and what the miscalibration measures. The predicted $p(z \in W \mid \mathbf{u}, Q)$ is low by a factor that depends on separation:

separation	n	mean predicted	observed	ratio
3–5''	674	0.038	0.368	9.8
5–10''	2,754	0.032	0.235	7.4
10–20''	9,912	0.025	0.123	4.9
20–30''	16,122	0.022	0.074	3.3

The scorer draws the companion’s redshift from the field distribution Σ_Q , but quasars cluster, so a companion at 3'' (about 25 kpc at $z \sim 1.5$) has an excess probability of sharing the primary’s redshift that the field prior does not contain. The shape of the prediction is right — the ranking says so — and its normalisation is short by the projected quasar–quasar correlation integrated

over the window. The excess is left out of the scorer on purpose, so that the ranking carries no cosmological assumption; a user who wants a probability multiplies the odds by the factor for their separation. The predecessor measured the same factors to two significant figures (10.0, 7.4, 4.9, 3.3), as it should: the ratio measures the sky, not the model. Whether it reaches unity beyond $30''$ is not yet measured.

10.3 Against the predecessor, on identical inputs

The predecessor (Appendix C) was validated on these same companions, so the two can be scored on exactly the same rows (`scripts/validate_pairs_multisurvey.py`). Its inputs were dereddened Legacy fluxes as ratios to r ; the model's are observed luminosities. Four configurations:

held-out AUC	g, r, z		$g, r, z, W1, W2$	
	predecessor	model	predecessor	model
same/wrong z by $\ln R$	0.703	0.700	0.815	0.828
same/wrong z by $p(z \in W Q)$	0.799	0.799	0.855	0.865
same/wrong z by $\ln \text{BF}$	0.667	0.655	0.759	0.720
same z against $3000\text{--}6000 \text{ km s}^{-1}$	0.498	0.494	0.488	0.472
quasar/non-quasar by $\ln \text{BF}$	0.890	0.891	0.970	0.966
quasar/star	0.903	0.874	0.988	0.982
quasar/galaxy	0.886	0.897	0.964	0.961

Table 2: The model against its predecessor on the same 14,221 held-out companions (880 same-redshift, 7,293 wrong-redshift and 6,048 non-quasar; the star and galaxy columns use all 5,295 and 15,989). The g, r, z columns mark $W1, W2$ unmeasured in the predecessor and supply only Legacy g, r, z to the model.

On identical g, r, z the two are equivalent in redshift discrimination (0.700 against 0.703 by $\ln R$); the model rejects stars less well (-0.029) and galaxies better ($+0.011$). With $W1, W2$ the model ranks redshift better ($+0.013$ by $\ln R$, $+0.010$ by the quasar-only probability) and rejects stars slightly less well (-0.006). Its $\ln R$ has Spearman correlation 0.940 with the predecessor's on these rows. Close ranking does not mean identical scores for every object.

The forced $W1, W2$ carry most of the discrimination: they add 0.11–0.13 in AUC to either model. The model's earlier version, trained without them because it held only AllWISE catalogue fluxes, matched the predecessor on g, r, z and could not use them; retraining with the Legacy forced bands as separate bands closed the gap.

10.4 Every survey combination

The pairs test Legacy photometry only. Every one of the 127 non-empty survey subsets is tested on reserved objects (`scripts/validate_multisurvey.py`): up to 200 reserved quasars and 200 reserved field sources per subset with at least one usable measurement from each requested survey and at least two in total, drawn from the reserved sky blocks and the reserved fields including the overlap field. The quasar/field test scores each quasar at its true redshift and each field source at a redshift drawn from those quasars, by colour Bayes factor; the redshift test scores each quasar at its true redshift and at a redshift drawn from the reserved quasars outside $\pm 2000 \text{ km s}^{-1}$, by the quasar-only redshift density under a flat prior. These are discrimination tests, distinct from the pairs, and they use neither surface densities nor U .

Over all 127 subsets the median quasar/field AUC is 0.996 and the minimum is VHS alone, 0.662: infrared-only input is scored, but the four VHS near-infrared bands alone barely separate quasars from the field. Combinations with SDSS u locate redshift best, as (13) predicts. The quasar-only redshift intervals are conservative: central 68% intervals contain the true redshift for a median 77% of quasars (126 of 127 subsets above nominal, range 0.66–0.88), and 95% intervals

survey subset	quasar/field AUC	true/other z AUC	68% coverage
SDSS	0.944	0.921	0.705
Legacy DR9 ($g, r, z, W1, W2$)	0.985	0.927	0.750
AllWISE	0.913	0.724	0.655
PS1	0.943	0.842	0.720
NSC	0.890	0.843	0.765
SkyMapper	0.839	0.794	0.685
VHS	0.662	0.754	0.740
Legacy + AllWISE	0.990	0.895	0.745
SDSS + AllWISE	0.987	0.948	0.820
AllWISE + VHS	0.931	0.842	0.770
all seven	1.000	0.981	0.790

Table 3: Reserved-object checks for representative survey subsets. Rows use different availability-selected populations and do not measure the gain from adding a survey to identical objects. The all-seven row has 145 field sources; the others 200 of each. Coverage is of central 68% redshift intervals under a flat prior, not calibration of the population posterior. All 127 rows are in `docs/MULTISURVEY_VALIDATION.md`.

for 96.5%. Against the previous 37-band version of the model, the Legacy-only row rises from 0.935 to 0.985 (quasar/field) and 0.886 to 0.927 (redshift), because Legacy now includes the forced $W1, W2$; single-survey rows otherwise move by up to 0.04, within the scatter of samples of 200.

11 Worked examples across survey combinations

Figure 8 is one of twelve versions of the same worked example: SDSS, Legacy, AllWISE, PS1, NSC, SkyMapper and VHS individually; Legacy+AllWISE, SDSS+AllWISE, PS1+AllWISE and AllWISE+VHS; and all seven. The full set, a twelve-page atlas, the ten objects’ photometry and all 120 score records are in `docs/examples/`. Every version uses the *same* five reserved quasars and five reserved field sources, drawn with seed zero before any score was computed, requiring the plotted bands in every survey and $19 < u_{\text{SDSS,r}} < 21.5$: 120 quasars in eight reserved blocks were eligible, one is drawn from each of five blocks, and the field sources all come from the reserved overlap field. The shared footprint and the coverage requirement make this an illustration, not a performance estimate.

Each panel projects the model onto two native luptitude colours conditioned on the measured reference band, with the plotted bands’ measurement covariance convolved before conditioning and every other band marginalised. Blue contours are the quasar density at z_0 , green dashed the field density; levels are fractions of peak density, not enclosed probabilities. Legacy-only input uses the dedicated Legacy field fit (§5.3); every other combination the joint one. The annotations use *every* band available in the combination, so the two plotted colours do not carry all the evidence.

input	Q1	Q2	Q3	Q4	Q5	F1	F2	F3	F4	F5
SDSS	+1.5	+5.6	+2.1	+0.2	+8.2	+0.5	−4.3	−3.4	−20.5	−2.0
Legacy	+5.2	+9.9	+7.3	+18.2	+11.5	−1.7	−10.7	−7.4	−38.5	−9.7
AllWISE	+9.7	+3.1	−1.2	+13.1	+3.5	+1.4	−1.0	−2.1	−5.1	−0.5
VHS	+5.0	+2.9	+4.4	+2.6	+2.8	+1.7	−1.8	−4.2	+0.9	+1.5
Legacy + AllWISE	+14.0	+9.6	+4.6	+15.8	+6.0	+0.7	−10.6	−9.6	−36.3	−9.7
all seven	+22.2	+30.8	+29.0	+37.8	+39.2	+6.9	−45.5	−36.4	−110.9	−14.0

$\ln \text{BF}$ at z_0 for each object and a selection of inputs. Q1–Q3 are at $z \approx 0.3$, Q4 at 1.14 and Q5 at 3.30; the field sources are assigned their column’s redshift and have no spectroscopic class. Evidence grows with the number of surveys, and Legacy alone now separates all five quasars from the field because it includes the forced $W1, W2$. F1 has positive evidence in most inputs, +6.9

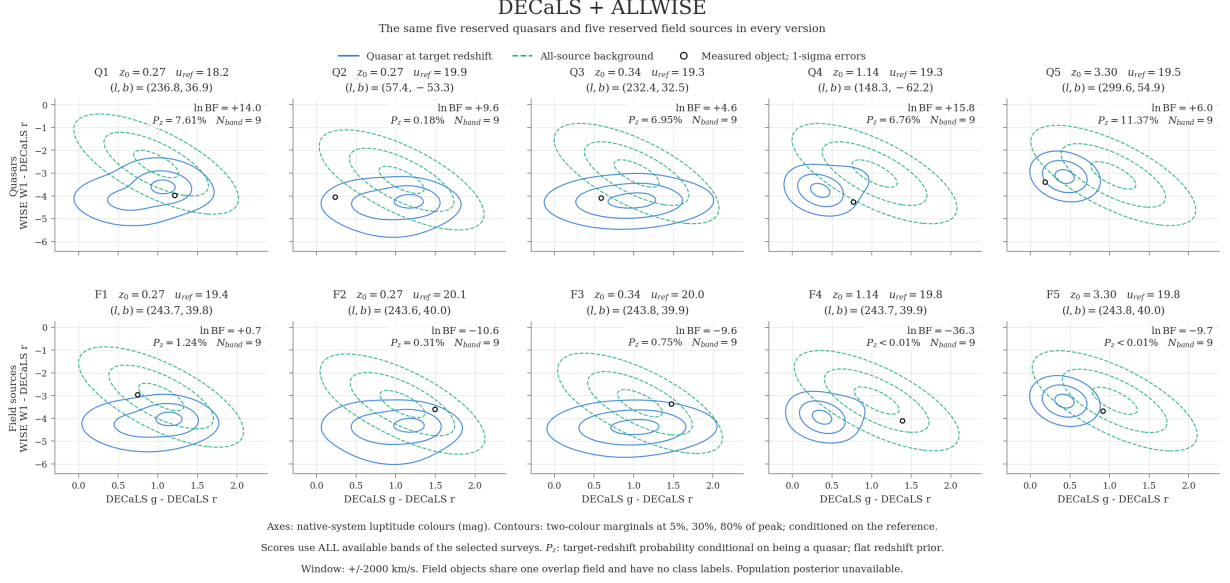


Figure 8: One of the twelve survey-combination versions: Legacy DR9 + AllWISE. The quasars Q1–Q5 (top) and field sources F1–F5 (bottom) recur in every version; each column shares one target redshift. Panel scores use all bands of the combination; P_z assumes the object is a quasar and is not p_{same} .

with all seven; neither it nor the weaker quasar cases were replaced to make the figure separate more cleanly.

12 Scope and limitations

Blending is the binding constraint. Below a few arcseconds the survey model fits divide flux between overlapping sources, so the two sets of photometry are neither independent nor individually reliable, and a probability computed from them describes the deblender rather than the sky. Every candidate scored for science passes an explicit separation floor and blending limit; a candidate lacking those measurements is treated as blended. Survey quality flags alone do not establish that a low-resolution infrared measurement — AllWISE in particular, with its $\sim 6''$ beam — separates a close pair; contaminated bands must be marked unusable. Closer pairs need image-level forced photometry.

p_{same} is not the probability of a physical pair. It is a classification posterior under field priors. A physical-pair calculation additionally conditions the quasar intensity on angular separation through small-scale quasar clustering (Hennawi et al., 2006; Eftekharzadeh et al., 2017). That layer is deliberately separate; its size is measured in §10.2, and a user who wants a probability multiplies the odds by that factor for their separation. The ranking statistic is unaffected.

Colour cannot resolve the velocity window. Quasars just outside $\pm 3000 \text{ km s}^{-1}$ are indistinguishable from those inside (§10.2). The ranking separates same-redshift quasars from the wrong-redshift population as a whole, and must not be read as velocity resolution.

The field model is coarse and global. Eight components over 41 bands, from 24 small fields at $|b| \geq 25^\circ$, with dedicated fits only for Legacy-only input. The predecessor measured that a local field model changes Legacy scores by at most 0.2 nats above $|b| = 20^\circ$ (Appendix C); nothing here tests the Galactic plane, and the surface densities are global.

Colours unlike anything either model has seen. The unmodelled hypothesis stops such objects being called certain quasars, at the cost of weaker Bayes factors for real quasars far from the field; it does not make the scorer informative there. An object flagged `outside_both_models` needs a spectrum or another look at its photometry.

Unused evidence. Gaia parallax and proper motion settle the question outright for bright companions. Variability, where it exists, is a further factor. The likelihood is modular so that such a term can be added as a factor.

A Gaussian identities used

For a mixture (9) over a partitioned vector $\mathbf{v} = [\mathbf{a}, \mathbf{b}]$ and an observation $\mathbf{b} = \mathbf{y}$ with covariance $\mathbf{S}_y$, the conditional is again a mixture, with weights

$$w_k \propto \alpha_k \mathcal{N}(\mathbf{y} \mid \boldsymbol{\mu}_{b,k}, \mathbf{V}_{bb,k} + \mathbf{S}_y), \quad (23)$$

and components

$$\boldsymbol{\mu}_{a|y,k} = \boldsymbol{\mu}_{a,k} + \mathbf{V}_{ab,k}(\mathbf{V}_{bb,k} + \mathbf{S}_y)^{-1}(\mathbf{y} - \boldsymbol{\mu}_{b,k}), \quad (24)$$

$$\mathbf{V}_{a|y,k} = \mathbf{V}_{aa,k} - \mathbf{V}_{ab,k}(\mathbf{V}_{bb,k} + \mathbf{S}_y)^{-1}\mathbf{V}_{ba,k}. \quad (25)$$

The conditional covariance is a Schur complement, and Schur complements preserve the Loewner order, which is what carries the dominance of the unmodelled density through conditioning (§8). Marginalising over a subset of dimensions deletes the corresponding rows and columns. No inverse is ever formed: every quadratic form and log determinant comes from a Cholesky factor, and all densities are computed in log space.

B The extreme-deconvolution update

For observation i with observed sub-vector $\mathbf{w}_i$, noise covariance $\mathbf{S}_i$, and component j , write O for the observed dimensions and

$$\mathbf{T}_{ij} = \mathbf{V}_j^{OO} + \mathbf{S}_i, \quad q_{ij} \propto \alpha_j \mathcal{N}(\mathbf{w}_i \mid \boldsymbol{\mu}_j^O, \mathbf{T}_{ij}), \quad (26)$$

$$\mathbf{b}_{ij} = \boldsymbol{\mu}_j + \mathbf{V}_j^O \mathbf{T}_{ij}^{-1}(\mathbf{w}_i - \boldsymbol{\mu}_j^O), \quad \mathbf{B}_{ij} = \mathbf{V}_j - \mathbf{V}_j^O \mathbf{T}_{ij}^{-1} \mathbf{V}_j^O, \quad (27)$$

with the M step

$$\alpha_j = \frac{1}{N} \sum_i q_{ij}, \quad \boldsymbol{\mu}_j = \frac{\sum_i q_{ij} \mathbf{b}_{ij}}{\sum_i q_{ij}}, \quad \mathbf{V}_j = \frac{\sum_i q_{ij} [\mathbf{B}_{ij} + (\mathbf{b}_{ij} - \boldsymbol{\mu}_j)(\mathbf{b}_{ij} - \boldsymbol{\mu}_j)^\top]}{\sum_i q_{ij}}. \quad (28)$$

With $\mathbf{S}_i = 0$ and no missing dimensions this reduces to ordinary Gaussian-mixture EM. Restricting O per object is how missing bands are handled. Sample weights multiply every sum; that is where the field sample's pattern weights enter. $\mathbf{T}_{ij}^{-1}$ appears only inside solves against a Cholesky factor.

C The predecessor model

The model replaces a Legacy-only one, retired on 2026-09-26 and kept in `models/archive/original_legacy_south/` because the model's own provenance depends on it: its sixteen reserved sky blocks are reserved in the model too, and its quasar prior fixes the model's completeness constant C (§6). `tests/test_archived_original.py` keeps its numbers reproducible.

What it was. Dereddened Legacy DR9 south g, r, z and forced $W1, W2$, as flux ratios f_b/f_r — the XDQSO representation (Bovy et al., 2011b), which keeps negative fluxes without a transform — with the Jacobian covariance $[C]_{jk} = \delta_{jk}v_j/f_r^2 + r_j r_k v_r/f_r^2$; 43 redshift slices of up to 20 components fitted to 1,106,986 quasars (917,489 DESI, 189,497 SDSS); a footprint-average field model, one mixture of eight components per r bin, fitted to all sources in eight 0.5° cones at $|b| > 32^\circ$; surface densities $\Sigma_Q(z, r)$ from DESI quasars normalised to the best-covered sky and $\Sigma_B(r)$ from those cones; and, from 2026-09-26, the same unmodelled hypothesis in its four flux-ratio coordinates.

What it measured, and still stands. On the pairs of §10, held-out AUC 0.815 by $\ln R$, 0.855 by $p(z \in W \mid Q)$, 0.759 by $\ln \text{BF}$, 0.488 against quasars just outside the window, 0.988 against stars and 0.964 against galaxies, with the clustering ratios 10.0, 7.4, 4.9, 3.3. It established several things this note relies on:

- *The field model must contain every source type.* Its first validation used a field fitted to point sources only and reported galaxies as a serious contaminant (AUC 0.80, 10.7% above the same-redshift median); an all-source field gave 0.96 and 0.3%.
- *Training volume does not set the redshift precision.* Fifteen times more quasars (1,237,336 against 79,865) and twice the components moved the median posterior width from 0.601 to 0.602.
- *A local field model hardly matters above $|b| = 20^\circ$.* In 30 cones from $|b| = 20^\circ$ to 78° , a field model fitted in the candidate’s own cone described held-out field sources better by at most 0.2 nats per object, left the false-positive rate unchanged, and moved quasar $\ln R$ by 0.1.
- *The colour tails decide far from both models (§8).*

Why it was retired. It took one survey. A second model took any combination of seven, but had no surface densities and so no R , and knew only AllWISE, not the Legacy forced $W1, W2$; on identical g, r, z the two were equivalent, and the original’s whole advantage came from those two bands. With reference-band priors (§6) and the forced bands added, the combined model matches or improves on the original on its own validation (Table 2), so there is no reason to keep two. The 37-band version of the combined model is archived beside it, with its priors and its initial joint-only version.

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